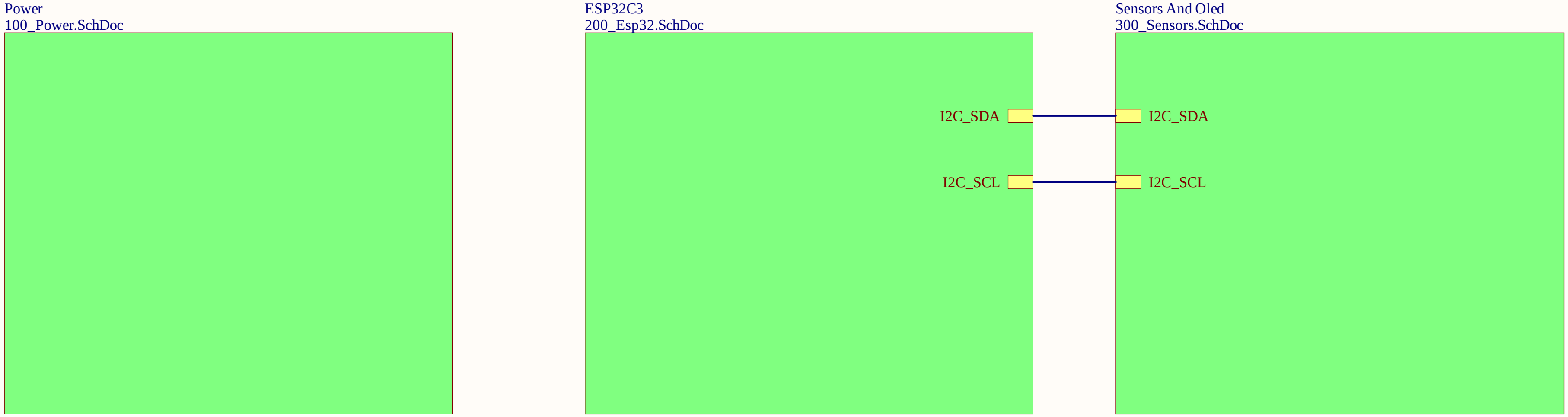

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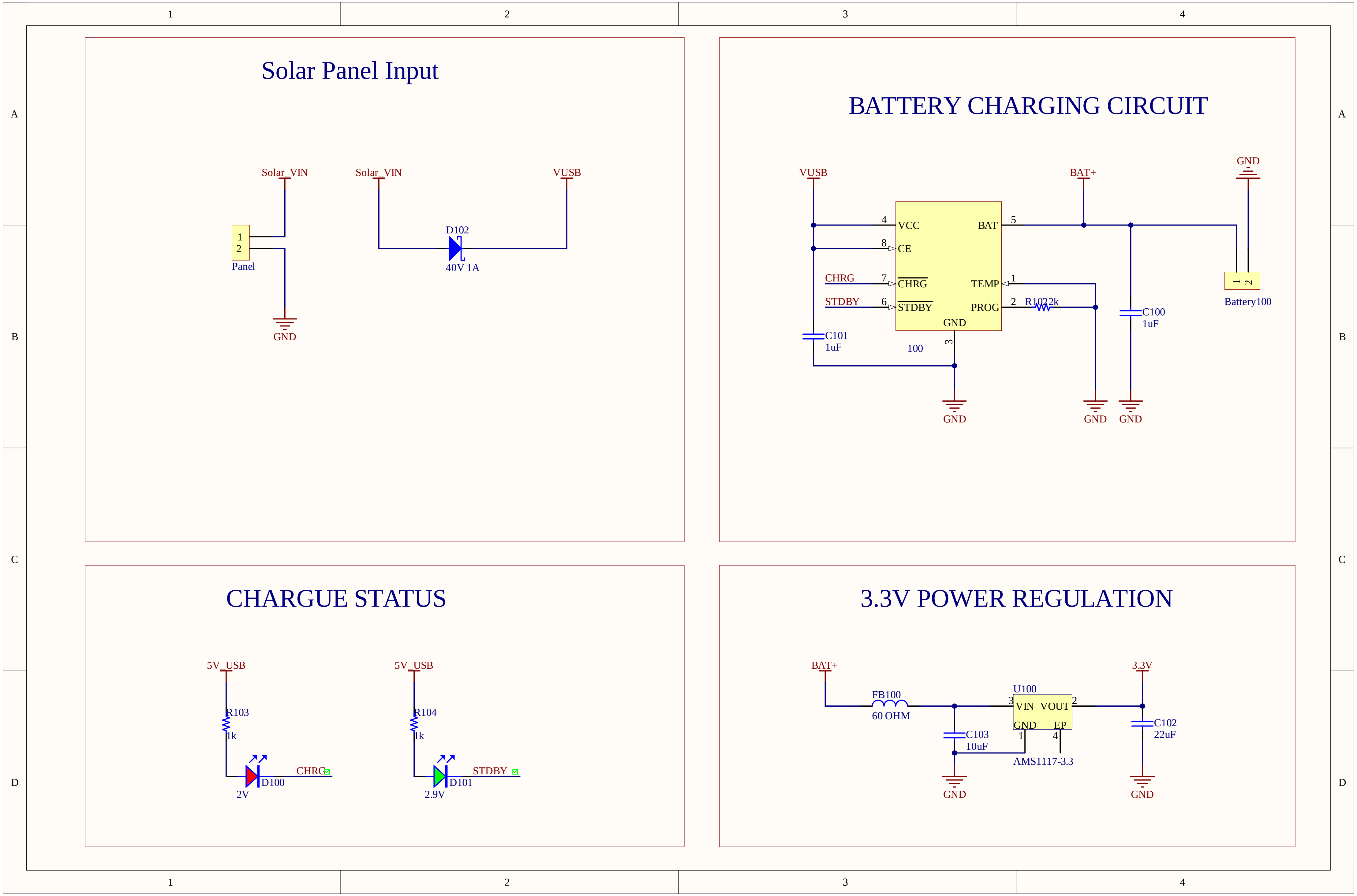
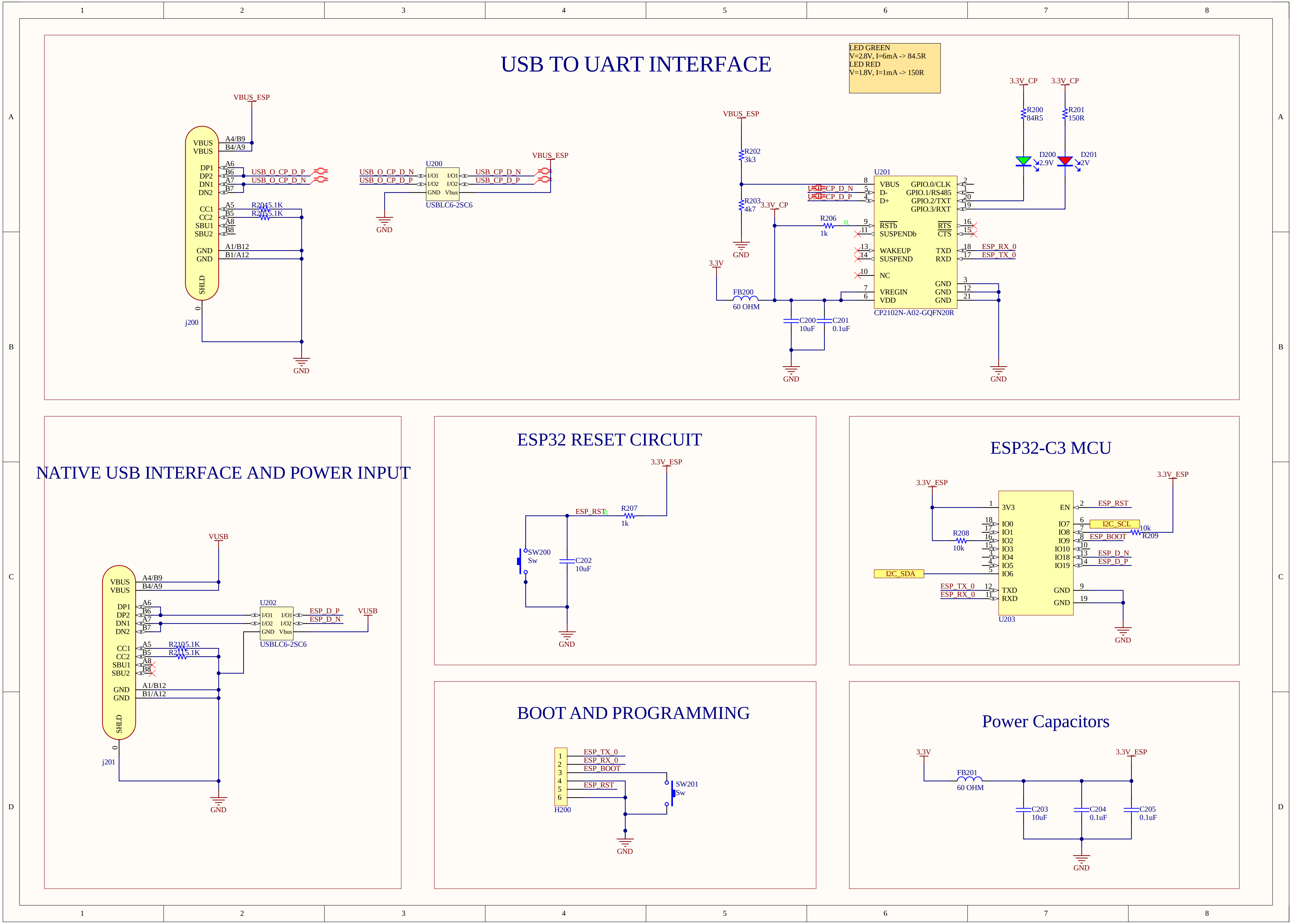
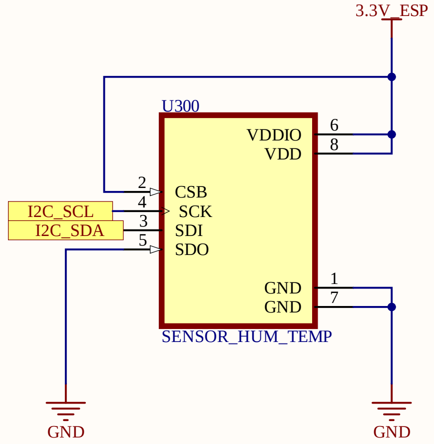

Solar Panel Input
Solar_VIN
Solar_VIN
VUSB
1
2
Panel
D102
40V 1A
GND
BATTERY CHARGING CIRCUIT
VUSB
BAT+
GND
VCC
BAT
CE
CHRG
TEMP
STDBY
PROG
GND
CHRG
STDBY
R102 2k
C100
1uF
C101
1uF
100
Battery100
GND
GND
GND
CHARGUE STATUS
5V_USB
5V_USB
R103
1k
R104
1k
CHRG
STDBY
D100
2V
D101
2.9V
3.3V POWER REGULATION
BAT+
3.3V
FB100
60 OHM
U100
VIN
VOUT
GND
EP
C103
10uF
C102
22uF
AMS1117-3.3
GND
GND
A
B
C
D
1
2
3
4

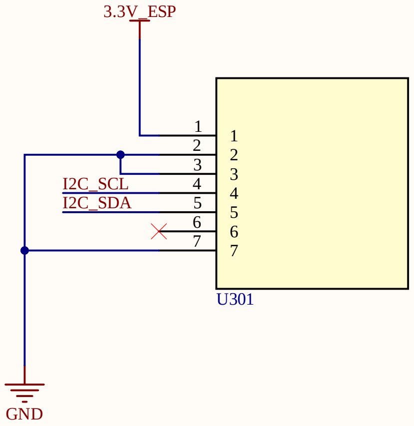


USB TO UART INTERFACE
LED GREEN
V=2.8V, I=6mA -> 84.5R
LED RED
V=1.8V, I=1mA -> 150R
VBUS_ESP
j200
U200
USBLC6-2SC6
USB_O_CP_D_P
USB_O_CP_D_N
USB_CP_D_N
USB_CP_D_P
GND
R202
3k3
R203
4k7
3.3V_CP
R206
1k
3.3V
FB200
60 OHM
C200
10uF
C201
0.1uF
U201
CP2102N-A02-GQFN20R
VBUS
D-
D+
RSTb
SUSPENDb
WAKEUP
SUSPEND
NC
VREGIN
VDD
GPIO.0/CLK
GPIO.1/RS485
GPIO.2/TXT
GPIO.3/RXT
RTS
CTS
TXD
RXD
ESP_RX_0
ESP_TX_0
R200
84R5
R201
150R
D200
2.9V
D201
2V
NATIVE USB INTERFACE AND POWER INPUT
VUSB
j201
U202
ESP_D_P
ESP_D_N
ESP32 RESET CIRCUIT
3.3V_ESP
ESP_RST
R207
1k
SW200
Sw
C202
10uF
ESP32-C3 MCU
U203
3V3
EN
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
TXD
RXD
R208
10k
R209
10k
I2C_SCL
I2C_SDA
ESP_BOOT
BOOT AND PROGRAMMING
H200
ESP_TX_0
ESP_RX_0
ESP_BOOT
ESP_RST
SW201
Sw
Power Capacitors
FB201
60 OHM
C203
10uF
C204
0.1uF
C205
0.1uF

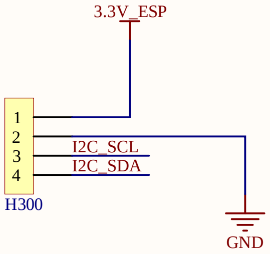
ENVIRONMENTAL SENSOR



LIGHT SENSOR



OLED DISPLAY HEADER



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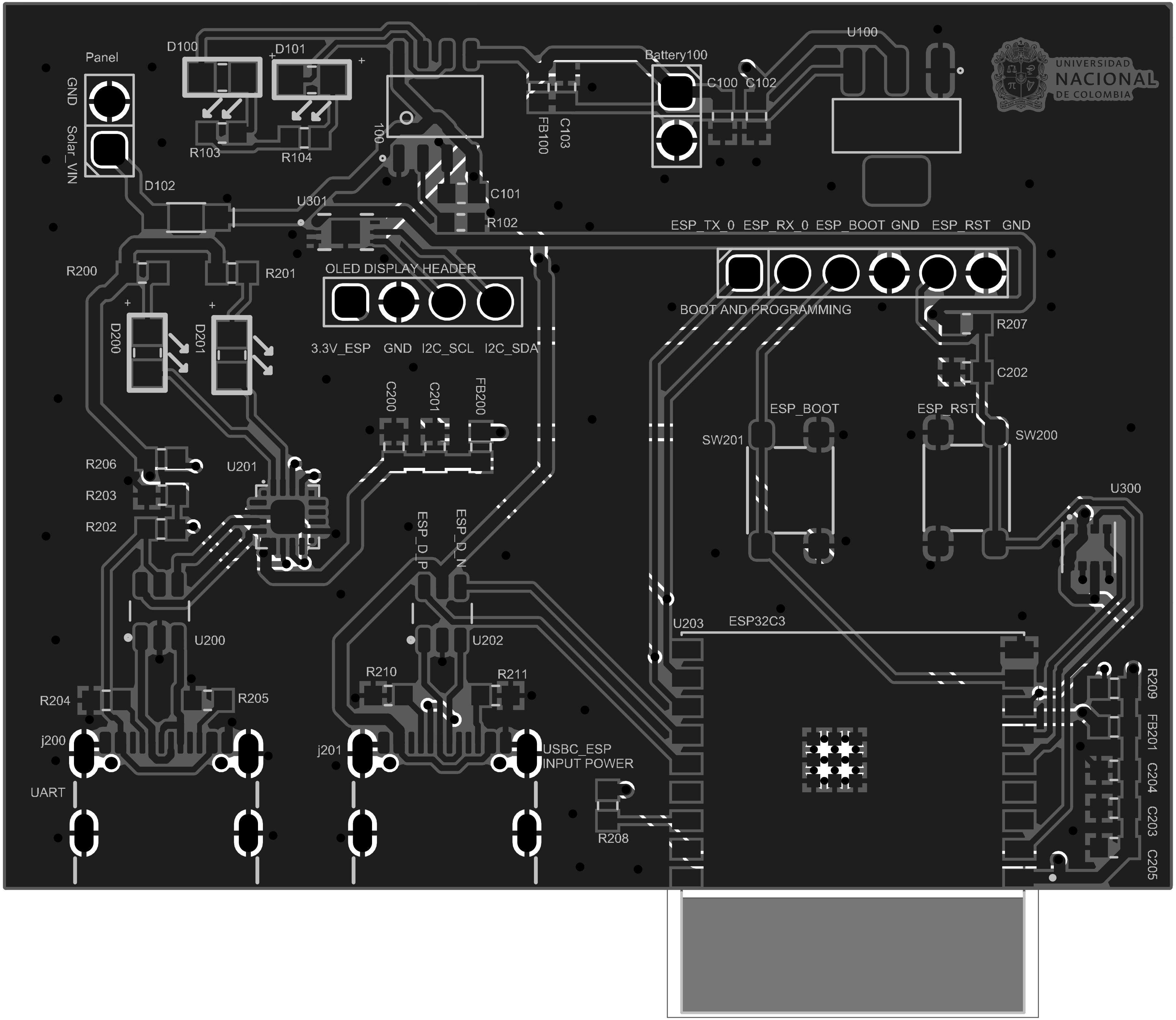
Panel
GND
Solar_VIN
D100
D101
R103
R104
D102
U301
C101
R102
FB100
C103
Battery100
C100
C102
U100
UNIVERSIDAD
NACIONAL
DE COLOMBIA
ESP_TX_0
ESP_RX_0
ESP_BOOT
GND
ESP_RST
GND
BOOT AND PROGRAMMING
R207
C202
ESP_BOOT
ESP_RST
SW201
SW200
R200
R201
OLED DISPLAY HEADER
3.3V_ESP
GND
I2C_SCL
I2C_SDA
D200
D201
C200
C201
FB200
R206
R203
R202
U201
ESP_D_P
ESP_D_N
U300
U203
ESP32C3
U200
U202
R210
R211
R204
R205
R209
FB201
C204
C203
C205
j200
j201
USBC_ESP
INPUT POWER
UART
R208